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Learning generalizable representations across Heterogeneous Acquisition Environments for Breast Ultrasound Diagnosis ☆

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ABSTRACT

Ultrasound image classification remains challenging in clinical practice, given substantial variations in image appearance across devices, acquisition protocols, and operator habits, often unrelated to underlying pathology and associated with inconsistent model performance and limited generalizability. To address this challenge, we propose **HAE-BUS (Heterogeneous Acquisition Environments for Breast Ultrasound**), a representation learning framework that aims to disentangle pathology-relevant features from acquisition-style bias and improve diagnostic consistency across heterogeneous acquisition conditions. Specifically, HAE-BUS uses Multimodal Large Language Models (MLLMs) as an offline semantic interface to characterize pathology-excluded acquisition cues, from which latent acquisition-style environments are inferred without relying on metadata. These inferred environments are then composed during training to suppress acquisition-style shortcuts while preserving pathology-relevant diagnostic features. Experiments conducted on two breast ultrasound benchmarks, including the public BUSBRA dataset and our private NDTH dataset, indicate that our approach obtains higher AUC and accuracy than the compared methods in the evaluated settings, along with more consistent performance across heterogeneous acquisition conditions, indicating improved robustness and suggesting potential for more reliable clinical translation.

1. Introduction

Breast cancer remains the most frequently diagnosed malignancy among women worldwide and one of the leading causes of cancer-related mortality (Bray et al., 2024). Early and accurate diagnosis is critical for improving survival outcomes. Ultrasound (US), with advantages of safety, affordability, and real-time imaging, has therefore become a frontline modality for breast cancer screening (Dang et al., 2025), lesion evaluation (Xu et al., 2024), and treatment monitoring (Conz et al., 2024), particularly in resource-limited or radiation-sensitive settings (Iacob et al., 2024; Dan et al., 2023).

Recent studies have further highlighted the biological heterogeneity of breast cancer and the potential of data-driven approaches for prognostic stratification and therapeutic target discovery (Shi et al., 2025). Advances in deep learning (Umirzakova et al., 2025), particularly convolutional neural networks (CNNs) (Liu et al., 2023) and Vision Transformers (Gheflati and Rivaz, 2022), have achieved superior in-domain performance in breast ultrasound analysis (Zhang et al., 2025). Beyond ultrasound classification, data-driven breast cancer studies have also explored tumor-cell subset identification, therapeutic target discovery, and outcome prediction (Yuan et al., 2026a,b, 2025). However, their performance degrades greatly under distribution shifts in clinical deployment (Zhang et al., 2020). Unlike CT or MRI, ultrasound acquisition is highly operator-dependent and sensitive to machine settings (Yi et al., 2024). Non-pathological factors, including device model, probe angle, scanning protocol, and operator experience (Crawford et al., 2023; Kripfgans et al., 2022; Yoon et al., 2011), introduce substantial style variability (brightness/contrast, background texture, and artifact patterns (Fetzer et al., 2022; Seoni et al., 2023; Kremkau and Taylor, 1986)) even for the same lesion type. From a causal perspective, such variability creates spurious correlations: models may latch onto background textures or device overlays and discount truly causal cues

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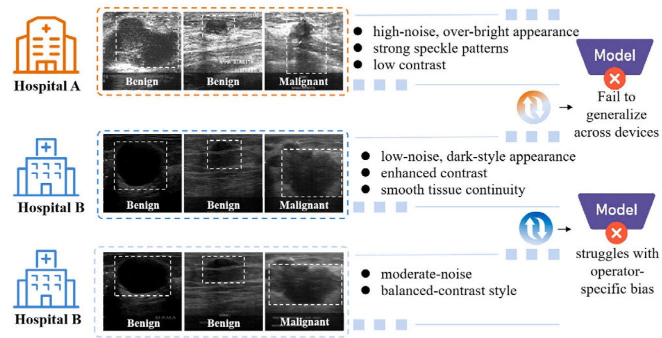


Fig. 1. Acquisition-style shift across hospitals and its impact on generalization. Each row shows benign and malignant examples from a different hospital, where lesions are similar but acquisition styles differ markedly. A model trained on a single-hospital style may fail to generalize to unseen styles, illustrating how out-of-distribution (OOD) performance degradation can be driven by style sensitivity rather than pathology.

(e.g., irregular margins, posterior acoustic attenuation). Consequently, out-of-distribution generalization deteriorates and clinical reliability is undermined (Jeffrey et al., 2025), as predictions hinge on acquisition style rather than pathology (Kilim et al., 2022; Song et al., 2024) (see Fig. 1).

To address these issues, one line of work focuses on data-level augmentation or style transfer with generative models (GANs, VAEs, diffusion models) (Kaleta et al., 2025; Lee et al., 2025). In spite of expanded diversity, they suffer from high training cost, imprecise style control, and the risk of lesion distortion (Saad et al., 2024). By contrast, we adopt a *model-level stable learning* perspective, which directly enforces *invariance* of predictive performance across heterogeneous environments (e.g., devices, operators, protocols). In this way, stable learning (Cui and Athey, 2022) drives the representation toward causal, pathology-relevant cues while suppressing style-induced shortcuts. Within this paradigm, *Invariant Risk Minimization (IRM)* (Arjovsky et al., 2019) offers a principled formulation: it searches for a representation Φ such that a single classifier w is optimal across all environments, aligning decision boundaries to lesion-specific evidence and mitigating spurious correlations. Compared with generative approaches, IRM is more computationally tractable and causally grounded, making it a compelling strategy for robust breast ultrasound classification.

Despite its causal appeal, the practical utility of IRM critically depends on high-quality environment partitions. In most medical datasets, metadata such as device type or acquisition site are either unavailable, inconsistently recorded, or insufficient to capture the nuances of style variability. Data-driven surrogates, such as unsupervised clustering or causal representation learning, offer limited interpretability and frequently disrupt class balance, undermining clinical applicability (Panayides et al., 2020). These limitations sharpen a single open question: how can we construct interpretable, pathology-excluding, and class-balanced environment partitions without relying on unreliable metadata?

Another limitation of conventional IRM methods lies in their uniform averaging of risks across environments $e \in \mathcal{E}$. In medical imaging, this static formulation faces two key drawbacks: (i) gradients from "easy" environments with low variability or noise dominate optimization, diluting signals from harder environments that more effectively expose spurious correlations; and (ii) the dynamic evolution of cross-environment relations during training cannot be captured by fixed uniform constraints. Thus, a *dynamic environment sampling* strategy is in urgent need, which adaptively prioritizes training batches maximizing cross-environment inconsistency e^*, thereby directing optimization toward the most fragile invariance breakpoints and steering the model away from style-driven shortcuts.

Recent advances in multimodal large language models (MLLMs) offer a promising direction. Beyond natural-language understanding, MLLMs exhibit strong cross-modal reasoning and instruction-following (Zhou et al., 2023), enabling the generation of *structured, pathology-excluded* descriptions of non-pathological style attributes (e.g., image quality, background texture, machine artifacts) from lesion-masked ultrasound inputs. These semantic style descriptors are interpretable, reproducible, and extensible, overcoming the subjectivity of manual annotation and the opacity of low-level clustering. When coupled with prototype-based reasoning, MLLMs can assign environment labels that are semantically grounded and *class-balanced*, establishing an auditable and clinically aligned foundation for *stable learning*.

Inspired by these observations, we develop *HAE-BUS*, an environment-guided representation learning framework for generalizable breast ultrasound classification under heterogeneous acquisition conditions. The main contributions are summarized as follows:

- We formulate generalizable breast ultrasound classification as representation learning across heterogeneous acquisition environments, where acquisition-induced variations are explicitly modeled as latent environments rather than treated as domain noise.
- We propose *HAE-BUS*, a metadata-free, environment-guided representation learning framework that derives acquisition-style partitions from pathology-excluded semantic cues generated by MLLMs.
- We introduce a cross-environment composition strategy that suppresses acquisition-style shortcuts and encourages pathology-relevant invariant representation learning across heterogeneous imaging conditions.

2. Related work

2.1. Multimodal LLM in ultrasound

Recent studies have extended multimodal large language models (MLLMs) to ultrasound, spanning quality assessment (B. Wang et al., 2025; Christensen et al., 2024; Maani et al., 2025), diagnosis (Zhou et al., 2025), and report generation (Yan et al., 2025). For standard-plane evaluation, (B. Wang et al., 2025) employ hierarchical prompting with LoRA tuning, whereas foundation-scale efforts such as Echo Interpretation (Christensen et al., 2024) and FetalCLIP (Maani et al., 2025) highlight the advantages of large-scale pretraining. UltraAD (Zhou et al., 2025) further exploits few-shot CLIP tuning for fine-grained anomaly classification. In diagnostic reporting, CLIP–GPT pipelines (Yan et al., 2025) exemplify the integration of visual representation and language modeling, advancing automated report generation in ultrasound.

Beyond task-specific architectures, general-purpose approaches have been introduced to enhance adaptability across diverse clinical tasks. MedBLIP (Gong et al., 2025) adapts image-language pretraining to medical visual question answering, while Med-SCoT (Qiao et al., 2025) emphasizes structured reasoning for interpretable medical VQA. Biomed-CoOp (Koleilat et al., 2025) and MedMoE (Chopra et al., 2025) integrate prompt learning and expert routing, while instruction-tuned assistants such as LLaVA-Ultra (Guo et al., 2024) and LLAUS (Guo et al., 2025) extend these capabilities to multi-task reasoning and interactive dialogue. Building on these advances, we propose to construct interpretable imaging style descriptions using MLLMs, thereby enabling stable learning under heterogeneous ultrasound conditions.

2.2. Stable learning

Stable learning has emerged from the intersection of causal inference and machine learning, with the goal of mitigating spurious correlations and improving robustness under distributional shifts (Cui and Athey, 2022; Athey et al., 2018). One line of work is grounded in covariate balancing, drawing on causal inference principles to control confounding and enhance generalization (Hainmueller, 2012; Athey et al., 2018; Chen et al., 2023). Zhang et al. (2021) introduced StableNet, which adaptively reweights training samples to suppress style-related variations while emphasizing causal features. In parallel to reweighting-based approaches, Heinze-Deml and Meinshausen (2021) formalized conditional variance penalties, demonstrating that explicitly disentangling core and style features further strengthens invariance. Xu et al. (2022) extended this line of work with a unified theoretical perspective based on independence-driven importance weighting.

Another line of work resorts to invariant risk minimization across domains (Arjovsky et al., 2019). This family of methods enforces predictors to remain simultaneously optimal under heterogeneous environments by penalizing environment-specific gradients or risks. Related medical domain generalization studies further improve robustness by synthesizing out-of-distribution appearances or adapting representations to unseen domains (M. Wang et al., 2025). Motivated by these, we adopt the stable learning paradigm to alleviate style-induced distribution shifts in ultrasound and to reinforce the extraction of pathology-centered causal features.

3. Preliminaries

3.1. Invariant risk minimization

Invariant Risk Minimization (IRM) is a principled framework that seeks to learn predictors generalizable across heterogeneous environments. Formally, IRM solves

$$\begin{aligned} &\min_{\Phi, w} \sum_{e\in\mathcal{E}} R^e(w\circ\Phi) \\ &\text{s.t.}\ \ w \in \arg\min_{\bar{w}} R^e(\bar{w}\circ\Phi)\,,\ \forall e\in\mathcal{E} \end{aligned} \tag{1}$$

where $R^e(\cdot)$ denotes the empirical risk in environment e. In this study, if Φ captures pathology-related causal features, then the same classifier w remains uniformly optimal across diverse imaging conditions.

However, due to its intractable bi-level optimization form, (Arjovsky et al., 2019) proposed a practical surrogate, IRMv1, which assumes that a fixed linear classifier $w = 1$ achieves the optimality:

$$\min_{\Phi, w} \sum_e R^e(w\circ\Phi) \ +\ \lambda\sum_e \left\|\nabla_{w|1.0} R^e(w\circ\Phi)\right\|^2 \tag{2}$$

where λ controls the strength of the invariance regularizer. This gradient norm penalty encourages the classifier $w = 1$ to be locally optimal and Φ capture causal features.

3.2. Theoretical intuition of cross-environment representation learning

As noted in Bae et al. (2021), Rosenfeld et al. (2020), effective invariant learning requires environments E_i that expose spurious variations while preserving stable pathology-relevant relationships. Ideally, the constructed environments should satisfy:

$$\begin{aligned} P(S_i \mid E_i = e_1) &\neq P(S_i \mid E_i = e_2), \\ P(Y_i \mid C_i, E_i = e_1) &= P(Y_i \mid C_i, E_i = e_2), \end{aligned} \quad \forall e_1, e_2 \in \mathcal{E}, \tag{3}$$

where C_i denotes pathology-related content and S_i denotes acquisition-related style factors. The first condition indicates that different environments should expose acquisition-style variations, whereas the second condition indicates that the pathology-label relationship should remain stable across environments.

To this end, *HAE-BUS* constructs acquisition environments from pathology-excluded semantic descriptors. The key intuition is that acquisition styles are usually implicit in pixel-level appearance and may be entangled with lesion morphology. As a result, direct clustering in the visual feature space may yield environments $\hat{E}_i = g_{\text{vis}}(X_i)$ that depend on both acquisition style S_i and pathology-related content C_i.

Table 1
Notations.

Symbol	Definition
$\mathcal{P}$	Prototype pool across environments
$\mathcal{P}_e$	Prototypes for environment e
B_{proto}	Number of style prototypes per environment
S_s	Small support set sampled from source environment E_s
Q_t^{probe}	Small probe set from target environment E_t
θ	Global model parameters before inner adaptation (shared across environments)
θ'_s	Source-specific adapted parameters after inner update on S_s
$\mathcal{L}^{\text{adv}}_{\text{qry}}(s)$	Adversarial query risk of $f_{\theta'_s}$ from E_s to sampled E_t
$\mathcal{L}_{\text{CE}}$	Cross-entropy classification loss
λ_1, λ_2	Weights for variability term (Std) and stability loss, respectively
N_{inner}	Number of inner adaptation steps on support set S_s

In contrast, a Multimodal Large Language Model (MLLM) serves as a pathology-controlled semantic interface that projects lesion-masked images $\tilde{X}_i$ into an acquisition-oriented descriptor space:

$$\begin{aligned} &D_i = F_{\text{MLLM}}(\tilde{X}_i; Q_{\text{style}}), \\ &I(D_i; S_i) > 0, \qquad I(D_i; C_i) \approx 0, \\ &E_i = \pi(D_i) \end{aligned} \tag{4}$$

Here, Q_{style} denotes the pathology-exclusion prompt, and $I(\cdot;\cdot)$ denotes mutual information. This formulation clarifies the intended role of the MLLM descriptors: they are not designed to introduce diagnostic knowledge, but rather to provide a semantic abstraction of acquisition heterogeneity, thereby encouraging the environment construction to focus on non-pathological style factors, such as image quality, contrast, background texture, and artifacts.

4. Methodology

In this section, we introduce in detail our proposed method, including Disentangled style description via MLLM (Section 4.2), Environment partitioning via instruction-guided reasoning (Section 4.3), Cross-environment task composition for stable learning (Section 4.4), the loss function (Section 4.5), and implementation details (Section 4.6).

4.1. Overview

We introduce **HAE-BUS (Heterogeneous Acquisition Environments for Breast Ultrasound)**, an MLLM-guided representation learning framework for robust breast ultrasound classification under acquisition-style shifts. Given a dataset $\mathcal{U} = \{(x_i, y_i)\}$, HAE-BUS derives K acquisition-style partitions $\mathcal{E} = \{E_1, E_2, \dots, E_K\}$ from pathology-excluded semantic cues generated by MLLMs, thereby providing structured environments for invariant representation learning. Cross-environment invariance constraints are then enforced to disentangle causal, pathology-relevant representations from spurious style factors.

As illustrated in Fig. 2, HAE-BUS consists of three modules: (1) **Disentangled Style Description** (Section 4.2): an MLLM generates structured style attributes that are independent of pathology. (2) **Instruction-Guided Environment Partitioning** (Section 4.3): these style descriptions are compared against prototypes to yield consistent environment labels via instruction-following. (3) **Cross-Environment Task Composition** (Section 4.4): the model is trained under simulated style shifts, combining source adaptation, adversarial environment sampling and stability regularization. This design encourages the learned features to remain stable across heterogeneous acquisition conditions, thereby enhancing both generalizability and interpretability in clinical ultrasound diagnosis. The symbols are defined in Table 1.

4.2. Structural style description via MLLM

Ultrasound images exhibit substantial variability arising from non-pathological factors such as device heterogeneity, acquisition protocols,

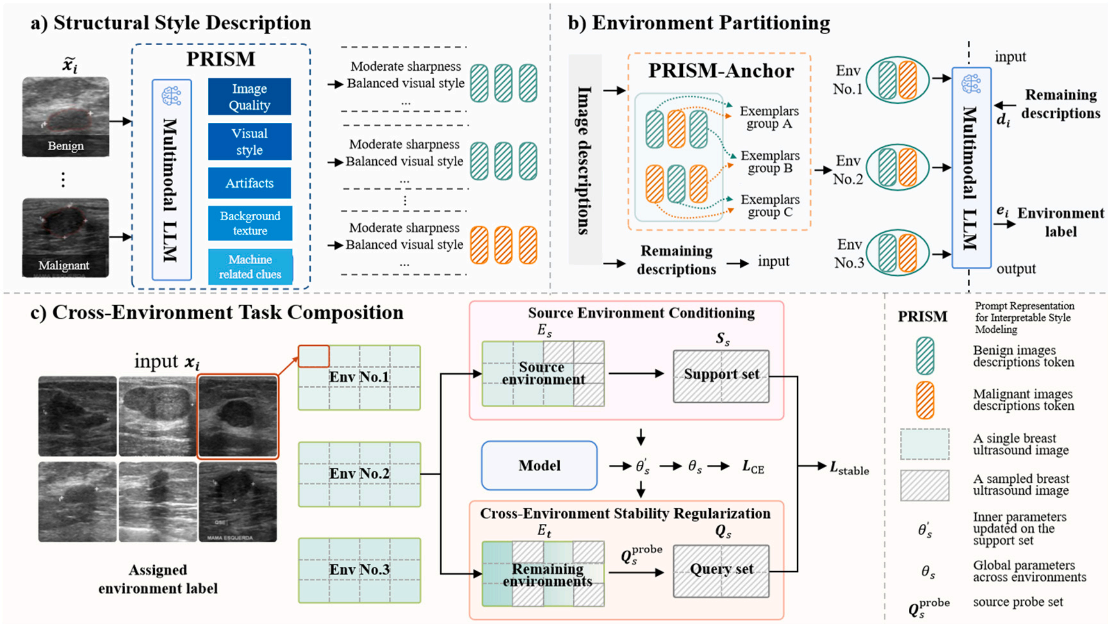


Fig. 2. Overview of HAE-BUS: MLLM-guided style reasoning, instruction-following environment partitioning, and cross-environment task composition. (a) Structural style description via MLLM. The PRISM prompt generates pathology-excluded style attributes (image quality, visual style, background texture, artifacts, machine-related clues) for each lesion-masked image. (b) Environment partitioning. Descriptions are compared with class-balanced style prototypes (PRISM-Anchor), and the MLLM assigns reproducible environment labels using PRISM-Guide. (c) Cross-environment task composition. The model adapts on a support set S_s from source environment E_s, then is evaluated on adversarial target environments E_t to compute query risks $\mathcal{L}^{\mathrm{adv}}_{\mathrm{qry}}(s \to t)$. The stability objective $\mathcal{L}_{\mathrm{stable}}$ regularizes cross-environment risk variability, enhancing out-of-distribution robustness in breast ultrasound diagnosis.

and operator-dependent habits. As discussed in Section 3, IRM seeks to mitigate the influence of such confounding factors by explicitly modeling distributional heterogeneity across environments. A key bottleneck of this framework is the availability of high-quality environment partitions; however, this issue has received little attention. To address this gap, we leverage the reasoning and cross-modal alignment capabilities of multimodal large language models (MLLMs) to derive structural ultrasound style descriptions, thereby enabling environment construction in a reproducible and interpretable manner.

First, each ultrasound image is preprocessed by overlaying the lesion mask as a contour, thereby preserving global context while explicitly delineating the lesion-background boundary. The resulting annotated images $\tilde{x}$ are subsequently provided to an MLLM under instruction-tuned prompts to extract style attributes such as acquisition quality, background texture, and operator-induced artifacts. The resulting standardized descriptions form a structured style set:

$$\mathcal{D} = \{d^{(1)}, d^{(2)}, \dots, d^{(M)}\}, \tag{5}$$

where each $d^{(m)}$ denotes a semantically consistent style descriptor, ensuring reproducibility and interpretability for environment construction in stable learning.

Then, we introduce **PRISM** (Prompt Representation for Interpretable Style Modeling), which specifies a unified prompt template $\mathcal{Q} = (\mathcal{S}, \mathcal{C}, \mathcal{J})$; each component plays a distinct role:

- $\mathcal{S} = \{s^{(m)}\}_{m=1}^{M}$ enumerates pathology-excluded **style dimensions**, such as `image quality`, `visual style`, `background texture`, `artifact presence`, and `machine signature`, etc.
- $\mathcal{C}$: **Content constraints** provide explicit instructions to suppress lesion-related information and prevent pathology leakage. For example, prompts are designed to highlight scanner parameters or visual noise patterns.
- $\mathcal{J}$: **JSON schema** defines a standardized, machine-readable output format where each style attribute is represented as a key–value pair. This structured representation facilitates automated parsing and seamless integration into environment construction pipelines.

Given the annotated input $\tilde{x}_i$, MLLM generates structured style description by cross-modal reasoning:

$$\mathcal{D}_i = \mathcal{F}_{\mathrm{M}}(\tilde{x}_i, \mathcal{Q}) = \{d_i^{(m)}\}_{m=1}^{M}, \quad d_i^{(m)} \in \mathbb{V}(s^{(m)}), \tag{6}$$

where $\mathcal{F}_{\mathrm{M}}$ denotes the ChatGPT-4o–based generator. Each $d_i^{(m)}$ is a typed element (categorical, ordinal, or numeric) with optional uncertainty. Guided by PRISM, the MLLM yields JSON-structured, pathology-excluded style descriptors covering image quality, background texture, artifacts, and machine-related clues, enabling reproducible environment construction.

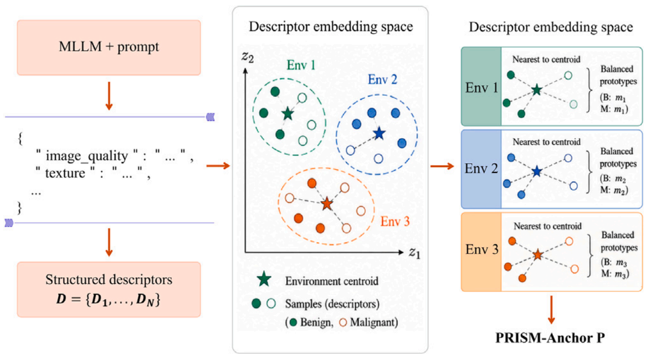
For representative style prototype selection, each generated descriptor is embedded into a semantic style space:

$$\mathbf{z}_i = f_{\mathrm{emb}}(\mathcal{D}_i), \tag{7}$$

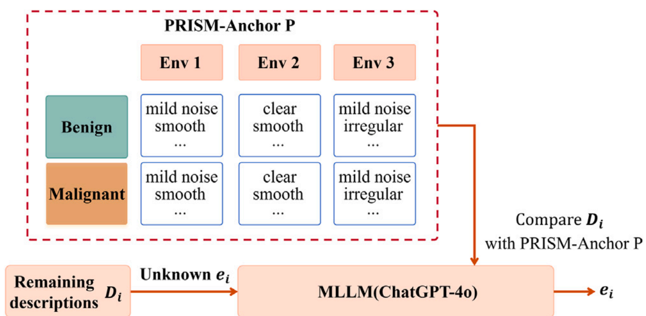
where f_{emb} denotes the descriptor embedding function. Based on $\{\mathbf{z}_i\}_{i=1}^{N}$, we estimate K representative style centers and retrieve the nearest real samples around each center as prototype candidates. These centers are used only to select representative anchors, rather than to directly assign environment labels. Class-balanced selection is further imposed around each center to reduce the risk of pathology-dominated prototypes. In this way, the MLLM-derived, pathology-excluded descriptors provide interpretable anchors for subsequent instruction-following environment partitioning, while avoiding the unstable or ambiguous prototype selection that may arise from conventional low-level feature clustering. Representative examples appear in the subsequent qualitative visualization (Fig. 12).

4.3. Instruction-following environment partitioning

Reliable environment partitioning is critical for stable learning, since noisy or biased labels can mislead models to capture spurious correlations rather than pathology-relevant features. To achieve



(a)



(b)

Fig. 3. Overview of MLLM-guided prototype-based environment construction. (a) Structured pathology-excluded descriptors are generated using a fixed MLLM prompt and embedded into a semantic style space. Prototype centers are used only to select centroid-nearest representative samples under benign/malignant class-balance constraints, forming environment-specific prototype anchors. (b) The selected prototypes serve as environment anchors, and each remaining descriptor is assigned to the most compatible environment by MLLM-guided style-matching scoring. The descriptors and environment labels are generated once during offline preprocessing and cached for downstream training.

reproducible and interpretable partitions, we propose an instruction-following framework that integrates balanced prototypes with MLLM-guided assignment.

Specifically, we construct a balanced prototype set, termed **PRISM-Anchor**, which serves as transparent anchors for subsequent reasoning:

$$\mathcal{P} = \{\mathcal{P}_1, \dots, \mathcal{P}_K\}, \quad \mathcal{P}_e = \{t_{e,1}, \dots, t_{e,B_{\mathrm{proto}}}\}, \tag{8}$$

where each $\mathcal{P}_e$ denotes pathology-excluded style exemplars (e.g., brightness, texture, artifacts). By explicitly enforcing balance across environments, PRISM-Anchor ensures diverse coverage of style factors and prevents partitions from being dominated by any single attribute.

PRISM-Guide denotes the instruction-following assignment rule that maps a description $\mathcal{D}_i$ to an environment label by comparing it with PRISM-Anchor.

For each candidate environment, the MLLM compares the query descriptor with the corresponding prototype group only along the pathology-excluded style dimensions defined in Section 4.2. This comparison produces a style-matching score that reflects the compatibility between the query descriptor and the acquisition-style pattern represented by each prototype group.

For an input $\tilde{x}_i$ with style description $\mathcal{D}_i$, the environment label is assigned as

$$e_i = \arg \max_{e \in \{1, \dots, K\}} p_{\mathrm{LLM}}(e \mid \mathcal{D}_i, \mathcal{P}), \tag{9}$$

where $p_{\mathrm{LLM}}(\cdot)$ denotes the style-matching score estimated by the MLLM through instruction-following comparison with PRISM-Anchor.

Diagnostic labels are not provided during this assignment stage, and each sample is assigned independently according to its highest compatibility score.

As illustrated in Fig. 3, each sample is processed independently to prevent label leakage and ensure reproducibility. The MLLM-based partitioning is performed once during offline preprocessing before model training; the generated descriptors and environment labels are cached and reused throughout training, and no MLLM query is required during inference.

The resulting environment set $\mathcal{E} = \{E_1, E_2, \dots, E_K\}$ is semantically interpretable, balanced across diagnostic categories, and reproducible across datasets. Compared with prior clustering-based approaches, our method (i) enforces balance by design, (ii) prevents pathology leakage through prototype auditing, and (iii) produces instruction-driven partitions rather than opaque feature clusters. Together with style descriptions (Section 4.2), this establishes a principled pipeline from semantic attribute extraction to robust environment construction.

4.4. Cross-environment task composition with adversarial sampling

With the initial environment partition, conventional IRM aggregates risks uniformly across environments, which often fails to expose the most fragile invariance breakpoints. This limitation is particularly pronounced in ultrasound imaging, where acquisition styles vary dynamically with operator habits, probe angles, and device settings, thereby violating the static-partition assumption underlying IRM. To address these limitations, we build on the meta-IRM paradigm (Bae et al., 2021) and extend it with the proposed *Cross-Environment Task Composition*, which actively simulates domain shifts via adversarial sampling.

Instead of uniformly enforcing invariance as in conventional IRM,

$$\min_{\theta} \sum_{E \in \mathcal{E}} \mathcal{L}(f_\theta; E), \tag{10}$$

we explicitly model cross-environment relations by sampling a *source environment* E_s and evaluating its generalization on *target environments* $E_t \in \mathcal{E} \setminus \{E_s\}$. Our method then adaptively emphasizes those source–target pairs that incur the largest meta-loss:

$$\min_{\theta} \ \mathbb{E}_{(E_s, E_t) \sim p_\psi} \tilde{\mathcal{L}}_{\mathrm{meta}}(s \to t), \tag{11}$$

thereby driving the model away from style-driven shortcuts. The procedure comprises three steps:

4.4.1. Source environment conditioning

At each iteration, one environment E_s is designated as the *source*, and the model is conditionally adapted on a small support set S_s:

$$\theta_s' \leftarrow \theta - \alpha \nabla_\theta \frac{1}{|S_s|} \sum_{(x,y) \in S_s} \mathcal{L}_{\mathrm{CE}}(f_\theta(x), y), \tag{12}$$

where α is the inner learning rate. This step simulates rapid within-environment adaptation, thereby reflecting the dynamic and heterogeneous acquisition conditions characteristic of ultrasound imaging.

4.4.2. Adversarial environment sampling

After conditioning on E_s, the adapted model $f_{\theta_s'}$ is evaluated on each of the remaining environments $E_t \in \mathcal{E} \setminus \{E_s\}$ using a small probe set Q_t^{probe}. This yields the meta-loss:

$$\tilde{\mathcal{L}}_{\mathrm{meta}}(s \to t) = \frac{1}{|Q_t^{\mathrm{probe}}|} \sum_{(x,y) \in Q_t^{\mathrm{probe}}} \mathcal{L}_{\mathrm{CE}}(f_{\theta_s'}(x), y). \tag{13}$$

Based on these estimates, we define an adversarial sampler that assigns higher probability to targets incurring larger meta-loss:

$$p_\psi(E_t \mid E_s) = \mathrm{softmax}\left(\frac{\tilde{\mathcal{L}}_{\mathrm{meta}}(s \to t) - b_s}{\tau}\right), \quad b_s = \frac{1}{|\mathcal{E} \setminus \{E_s\}|} \sum_t \tilde{\mathcal{L}}_{\mathrm{meta}}(s \to t). \tag{14}$$

Here, τ is a temperature parameter controlling distribution sharpness, and b_s is a per-source baseline that stabilizes sampling across environments.

4.4.3. Cross-environment stability regularization

We next evaluate the adapted model $f_{\theta'_s}$ on adversarially sampled target environments. The corresponding query risk is defined as

$$\mathcal{L}_{\text{qry}}^{\text{adv}}(s) = \mathbb{E}_{E_t \sim p_\psi(\cdot|E_s)} \left[\frac{1}{|Q_t|} \sum_{(x,y)\in Q_t} \mathcal{L}_{\text{CE}}(f_{\theta'_s}(x), y) \right], \tag{15}$$

which simulates worst-case cross-environment transfer. If the predictor f_θ relies on spurious style correlations, performance will degrade most significantly on these adversarially sampled targets.

To enforce stability, we aggregate the adversarial query risks across all source–target pairs and constrain both their mean and variability:

$$\mathcal{L}_{\text{stable}} = \frac{1}{|\mathcal{E}|} \sum_{E_s\in\mathcal{E}} \mathcal{L}_{\text{qry}}^{\text{adv}}(s) + \lambda_1 \cdot \text{Std}\Big(\{\mathcal{L}_{\text{qry}}^{\text{adv}}(s\to t)\}_{s,t} \Big). \tag{16}$$

The first term penalizes high average risk, while the second discourages large cross-environment variance, thereby promoting consistent performance across heterogeneous conditions.

In contrast to conventional IRM, which passively aligns risks over static environments, our approach *actively identifies and regularizes the most adversarial style shifts*. This design offers two key benefits: (i) improved robustness under dynamic and heterogeneous ultrasound acquisitions, and (ii) stronger disentanglement of pathology-relevant features from acquisition-related artifacts.

4.5. Loss function

The proposed objective integrates standard classification loss with the stability regularizer. Specifically, the cross-entropy loss over a mini-batch of size B is defined as

$$\mathcal{L}_{\text{CE}} = -\frac{1}{B} \sum_{i=1}^{B} \sum_{c=1}^{C} y_{i,c} \log p_{i,c}, \tag{17}$$

where C denotes the number of classes, $y_{i,c}$ the one-hot label, and $p_{i,c}$ the predicted probability for class c.

To promote robustness under heterogeneous environments, we further introduce the stability loss $\mathcal{L}_{\text{stable}}$ (Eq. (16)), which penalizes both elevated mean query risk and large variance across environment pairs. The overall optimization objective is therefore formulated as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda_2\, \mathcal{L}_{\text{stable}}, \tag{18}$$

where λ_2 is a trade-off parameter controlling the balance between classification accuracy and cross-environment stability.

4.6. Implementation details

All experiments were conducted on an NVIDIA RTX 4090 GPU with PyTorch 2.1. The backbone is a ResNet-50 pretrained on ImageNet, followed by a task-specific classifier. For environment construction, we set the number of environments to $K = 3$ with $B_{\text{proto}} = 3$ style prototypes per environment. The MLLM (ChatGPT-4o) generated style descriptions under a fixed PRISM prompt with decoding temperature set to 0.0 to reduce output randomness.

For classifier training, ultrasound images were resized to 224×224 and intensity-normalized before being fed into the backbone. Lesion masks were used only during the offline MLLM-based descriptor-generation stage to generate pathology-excluded style descriptions and environment labels. We used Adam with learning rate $\eta = 1 \times 10^{-4}$ and inner adaptation step size $\alpha = 1 \times 10^{-3}$; batch size $B = 32$. Hyperparameters were tuned on a validation split: $\lambda_1 = 0.1$, $\lambda_2 = 0.5$, and $N_{\text{inner}} = 3$. Early stopping was applied if validation performance did not improve for 20 epochs.

For reproducibility, we provide the complete prompt design, representative input–output examples, structured JSON schema, pathology-exclusion constraints, and the offline descriptor-generation/caching protocol in the **Supplementary Section S2**.

5. Experiments

In this section, we first describe the datasets, experimental setup, and competing methods. We then evaluate the proposed framework in terms of overall diagnostic performance on BUSBRA and NDTH, followed by cross-site external validation and leave-one-device-out evaluation to assess robustness under site/cohort-level and device-level acquisition shifts. Next, we analyze held-out generalization under different environment partitioning strategies and examine the influence of environment partitioning on ERM and DRO. We further report ablation studies to evaluate the contributions of PRISM-based environment partitioning and cross-environment task composition. In addition, we investigate the effects of the number of environments and the dimensionality of style descriptions, followed by prompt sensitivity analysis, repeated-run consistency evaluation, and pathology-leakage analysis. Finally, we present qualitative visualization to illustrate the semantic coherence and acquisition-style characteristics of the inferred environments.

5.1. Datasets and experimental setup

We evaluate the proposed method on one public and one private breast ultrasound dataset: **BUSBRA** (Gómez-Flores et al., 2024) (public) and **NDTH** (private, collected at *Nanjing Drum Tower Hospital*). To ensure fair assessment, each dataset is randomly split into training (60%), validation (20%), and test (20%) sets with class balance preserved. This stratified random splitting was repeated five times with different random seeds, and the same five splits were used for all compared methods. Hyperparameters and early stopping were determined only on the validation set, and the test set was used exclusively for final evaluation. We report the mean and standard deviation across the five repeated runs. Diagnostic performance is measured by Accuracy (ACC), Area Under the ROC Curve (AUC), Sensitivity (SEN), Specificity (SPE), Precision, and F1-score.

BUSBRA dataset

BUSBRA is a public breast ultrasound dataset from the Brazilian National Cancer Institute, containing **1,875** images from **1,064** patients (722 benign, 342 malignant; all pathology-confirmed and expert-annotated). Images were acquired on multiple devices, introducing substantial style heterogeneity. We use all left-side (`-l`) and sagittal (`-s`) images, totaling **1,064** samples.

NDTH dataset

NDTH is a private breast ultrasound dataset collected at *Nanjing Drum Tower Hospital*, comprising 1025 pathology-confirmed cases verified by biopsy or surgery (372 benign, 653 malignant). Images were acquired on a Philips iU22 system under routine clinical protocols, providing a representative spectrum of acquisition conditions and lesion appearances suitable for robust evaluation.

5.2. Competing methods

To evaluate the effectiveness of the proposed *HAE-BUS* framework, we compare it with state-of-the-art methods across four categories: conventional CNN ensembles, diffusion-based models, Mamba-based models, and domain generalization/invariant learning methods:

- **Fuzzy-Ensemble** (Manna et al., 2021): Fuzzy rank-based ensemble of multiple CNNs for robust classification under class imbalance.
- **BiomedCLIP** (Zhang et al., 2023): Contrastive image–text pretraining on large biomedical corpora for strong transfer.
- **DiffMIC** (Yang et al., 2023): Diffusion-based classifier combining global context and lesion-aware priors.

Table 2
Diagnostic performances comparison of various methods. The terms a and b in "a ± b" denote the mean and standard deviation (std.). * indicates $p < 0.05$, and ** indicates $p < 0.01$, based on the Wilcoxon signed-rank test comparing the best-performing method with the second-best method.

Method	AUC	ACC	SEN	SPE	F1-score	Precision
			Differentiation of BUSBRA Breast Dataset			
Fuzzy-Ensemble (Manna et al., 2021)	81.91 ± 3.12	78.13 ± 2.21	81.10 ± 3.58	76.72 ± 3.21	77.60 ± 2.60	80.61 ± 2.43
BiomedCLIP (Zhang et al., 2023)	80.67 ± 3.08	81.50 ± 1.46	80.42 ± 2.76	82.01 ± 2.73	80.25 ± 2.69	83.63 ± 2.68
DiffMIC (Yang et al., 2023)	87.48 ± 1.95	84.38 ± 1.05	71.00 ± 6.25	90.70 ± 2.71	72.88 ± 2.56	83.92 ± 5.35
BU-Mamba (Nasiri-Sarvi et al., 2024)	89.06 ± 1.41	87.35 ± 0.89	**84.66 ± 1.58**	89.21 ± 0.99	84.32 ± 1.66	86.91 ± 1.34
StableNet (Zhang et al., 2021)	85.94 ± 1.60	82.92 ± 1.29	70.26 ± 4.90	88.92 ± 4.16	78.74 ± 1.02	83.51 ± 3.36
IRMv1(Arjovsky et al., 2019)	87.56 ± 1.34	86.94 ± 2.17	82.94 ± 3.64	88.27 ± 1.54	**84.78 ± 1.78**	**86.95 ± 1.20**
MixStyle (Zhou et al., 2021)	88.81 ± 0.34	83.57 ± 0.57	74.57 ± 2.16	87.83 ± 1.41	74.45 ± 0.84	74.41 ± 1.95
VREx (Krueger et al., 2021)	88.35 ± 0.47	82.72 ± 1.74	74.63 ± 2.65	86.56 ± 3.27	73.60 ± 1.64	72.80 ± 4.09
PLDG (Yan et al., 2024)	90.27 ± 0.22	85.35 ± 0.90	74.34 ± 3.26	90.58 ± 2.07	76.57 ± 1.31	79.13 ± 3.11
CFFDL (Li et al., 2024)	89.66 ± 1.29	84.88 ± 1.42	75.53 ± 5.87	89.34 ± 2.88	76.24 ± 2.42	77.39 ± 3.96
HAE-BUS(Ours)	**91.79 ± 1.05****	**87.88 ± 0.95****	80.78 ± 5.62	**91.19 ± 2.31***	80.89 ± 1.91	81.35 ± 3.36
			Differentiation of NDTH Breast Dataset			
Fuzzy-Ensemble (Manna et al., 2021)	90.72 ± 1.80	84.82 ± 4.19	76.79 ± 6.36	92.86 ± 10.24	84.72 ± 4.40	90.74 ± 3.12
BiomedCLIP (Zhang et al., 2023)	93.85 ± 0.51	87.73 ± 0.41	92.16 ± 1.78	77.86 ± 3.68	85.45 ± 0.51	86.10 ± 0.97
DiffMIC (Yang et al., 2023)	90.17 ± 2.16	76.98 ± 2.32	66.87 ± 4.60	**94.86 ± 4.91**	78.71 ± 2.79	**94.88 ± 2.58**
BU-Mamba (Nasiri-Sarvi et al., 2024)	93.60 ± 0.60	90.17 ± 1.52	91.93 ± 1.57	86.17 ± 5.02	89.26 ± 1.84	89.62 ± 1.64
StableNet (Zhang et al., 2021)	90.74 ± 1.39	82.32 ± 0.78	90.93 ± 2.47	63.10 ± 3.04	78.24 ± 0.37	80.41 ± 1.90
IRMv1(Arjovsky et al., 2019)	93.74 ± 0.84	87.02 ± 2.60	84.54 ± 3.70	89.70 ± 3.70	85.03 ± 2.93	82.43 ± 4.18
MixStyle (Zhou et al., 2021)	90.34 ± 2.26	83.00 ± 3.12	87.47 ± 7.94	72.97 ± 8.11	87.54 ± 3.03	88.11 ± 2.47
VREx (Krueger et al., 2021)	90.99 ± 1.61	82.00 ± 2.35	78.80 ± 4.54	89.19 ± 3.94	85.77 ± 2.22	94.32 ± 1.73
PLDG (Yan et al., 2024)	92.44 ± 2.37	83.58 ± 2.37	94.94 ± 2.65	58.11 ± 8.33	88.89 ± 1.50	83.65 ± 2.55
CFFDL (Li et al., 2024)	93.05 ± 1.63	84.00 ± 3.29	82.17 ± 4.75	88.11 ± 3.08	87.61 ± 2.76	93.95 ± 1.50
HAE-BUS(Ours)	**95.44 ± 1.11***	**93.26 ± 0.91***	**96.16 ± 2.79***	86.79 ± 5.73	**95.17 ± 0.69**	94.28 ± 2.79

- **BU-Mamba** (Nasiri-Sarvi et al., 2024): Mamba-based vision encoder that fuses CNN locality with long-range dependencies.
- **StableNet** (Zhang et al., 2021): Stable-learning method that reweights samples to downweight style-related variations.
- **IRMv1** (Arjovsky et al., 2019): Invariant risk minimization baseline that penalizes environment-wise gradients to enforce invariance.
- **MixStyle** (Zhou et al., 2021): a feature-level style augmentation method that mixes instance-level feature statistics.
- **VREx** (Krueger et al., 2021): a risk-variance regularization method that encourages consistent risks across environments.
- **PLDG** (Yan et al., 2024): a prompt-driven latent domain generalization method for medical image classification.
- **CFFDL** (Li et al., 2024): a causal feature decomposition method that learns domain-invariant task-relevant representations.

5.3. Comparison on diagnostic performance

We compare the performance of HAE-BUS with ten state-of-the-art breast ultrasound classification methods on two datasets—BUSBRA (public) and NDTH (private). Quantitative results are reported in Table 2.

On the BUSBRA dataset, HAE-BUS achieves the highest AUC of 91.79% and the best accuracy of 87.88%, together with the highest specificity of 91.19%. BU-Mamba yields the highest sensitivity of 84.66%, whereas IRMv1 obtains the best F1-score and precision of 84.78% and 86.95%, respectively. Among the representative domain generalization methods, PLDG achieves the strongest AUC and ACC of 90.27% and 85.35%, respectively, while CFFDL obtains the highest sensitivity of 75.53%. In comparison, HAE-BUS reaches 91.79% AUC, 87.88% ACC, and 80.78% sensitivity, yielding higher values than the corresponding DG baselines. For instance, DiffMIC and StableNet show relatively low sensitivity of 71.00% and 70.26%, indicating a pronounced sensitivity-specificity mismatch. In contrast, HAE-BUS tends to favor overall discrimination and specificity under the current threshold, albeit with slightly lower sensitivity of 80.78% and F1-score of 80.89%. Although our method achieves the highest AUC and accuracy on BUSBRA, its threshold-dependent metrics, including sensitivity, F1-score, and precision, are not the best among all methods. This reflects a tradeoff under the current decision threshold, where the model favors stronger overall discrimination and specificity and may be useful in reducing unnecessary biopsies. In clinical practice, a sensitivity-prioritized threshold (lower) may be preferable when the application prioritizes cancer screening or malignant-case recall.

On the NDTH dataset, HAE-BUS attains the best overall performance, achieving an AUC of 95.44% and an accuracy of 93.26%, along with a sensitivity of 96.16% and an F1-score of 95.17%. Among the representative domain generalization methods, CFFDL achieves the highest AUC and ACC of 93.05% and 84.00%, respectively, while PLDG obtains the highest sensitivity and F1-score of 94.94% and 88.89%. Compared with these DG results, HAE-BUS reaches higher AUC, ACC, sensitivity, and F1-score, while maintaining comparable specificity and precision. BU-Mamba provides the closest results among the conventional classification baselines, with an AUC of 93.60% and an accuracy of 90.17%, whereas BiomedCLIP exhibits a clearer sensitivity-specificity trade-off. DiffMIC reports high specificity and precision (94.86% and 94.88%), yet its sensitivity and accuracy remain lower at 66.87% and 76.98%. These results suggest that cross-environment task composition contributes to reducing the influence of style-related variations and improving diagnostic robustness under the evaluated setting.

Overall, HAE-BUS achieves favorable performance on both BUSBRA and NDTH, with the highest AUC and ACC on the two datasets. On BUSBRA, it mainly shows advantages in overall discrimination and specificity, while on NDTH it achieves higher sensitivity and F1-score together with strong overall accuracy. Compared with representative domain generalization methods, HAE-BUS shows favorable overall discrimination, while threshold-dependent metrics vary across datasets and baselines. These results suggest that constructing pathology-excluded acquisition environments can help the model reduce its dependence on dataset-specific imaging styles and improve robustness across heterogeneous breast ultrasound data.

5.4. Cross-site external validation

Specifically, two cross-site transfer settings were considered: NDTH→BUSBRA and BUSBRA→NDTH, where each model was trained on the source cohort/site and directly evaluated on the independent target cohort/site.

Table 3
Performance comparison of different methods in cross-site external validation.

Method	AUC	ACC	SEN	SPE	AUC Gap↓
			Training: NDTH, Testing: BUSBRA		
MixStyle	79.30 ± 0.69	71.79 ± 1.18	68.89 ± 5.63	73.51 ± 5.08	9.51 ± 0.41
VREx	78.50 ± 0.50	72.03 ± 1.16	67.59 ± 6.86	74.68 ± 5.86	9.85 ± 0.21
IRMv1	77.35 ± 0.85	71.88 ± 2.15	65.43 ± 4.64	76.02 ± 5.51	10.21 ± 0.87
PLDG	78.24 ± 1.61	70.39 ± 5.52	69.45 ± 9.11	70.96 ± 14.01	12.03 ± 1.42
CFFDL	79.73 ± 1.08	74.91 ± 1.54	65.98 ± 2.69	**80.24 ± 3.41**	9.92 ± 0.58
HAE-BUS (Ours)	**84.95 ± 1.55**	**77.46 ± 2.07**	**75.17 ± 4.96**	79.20 ± 6.32	**6.84 ± 0.69**
			Training: BUSBRA, Testing: NDTH		
MixStyle	82.11 ± 2.91	74.21 ± 2.67	73.40 ± 4.43	75.96 ± 2.88	8.24 ± 1.54
VREx	84.72 ± 1.32	78.60 ± 1.61	**84.54 ± 4.54**	66.29 ± 5.95	6.28 ± 1.08
IRMv1	82.62 ± 2.07	73.03 ± 1.34	67.84 ± 3.05	83.79 ± 5.19	11.12 ± 1.37
PLDG	83.90 ± 1.37	75.19 ± 2.02	72.65 ± 5.67	80.46 ± 5.90	8.54 ± 1.42
CFFDL	81.91 ± 1.14	72.76 ± 1.33	70.77 ± 4.92	77.20 ± 8.24	11.15 ± 0.91
HAE-BUS (Ours)	**91.05 ± 2.16**	**83.81 ± 2.33**	83.20 ± 2.65	**85.03 ± 3.36**	**4.39 ± 1.35**

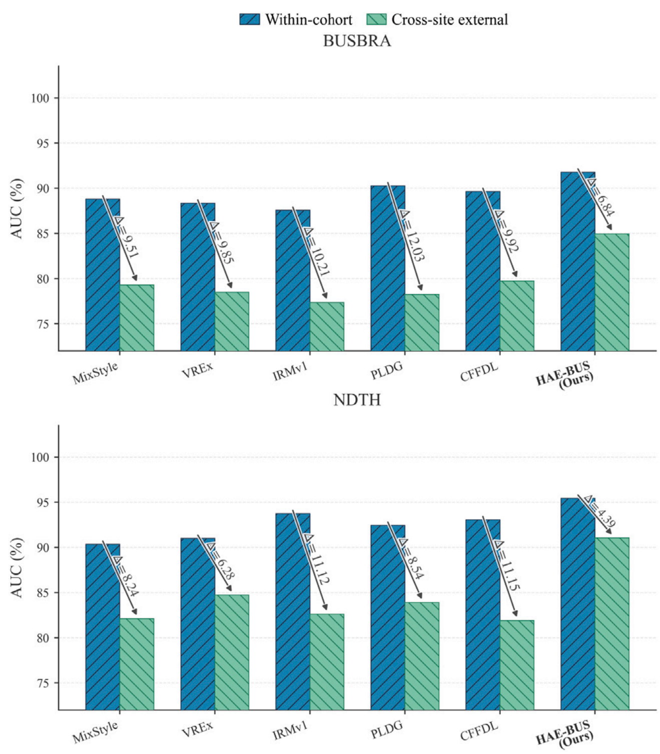


Fig. 4. Comparison of AUC between within-cohort and cross-site external evaluations.

As shown in Table 3, HAE-BUS achieves the highest AUC and ACC and the smallest AUC Gap in both transfer directions. In the NDTH→BUSBRA setting, HAE-BUS obtains an AUC of 84.95% and an ACC of 77.46%. In the BUSBRA→NDTH setting, HAE-BUS achieves an AUC of 91.05% and an ACC of 83.81%. These results indicate that the proposed environment-guided representation learning strategy improves diagnostic generalization under site-level acquisition shifts.

We further used the AUC Gap to quantify the performance decrease from within-cohort evaluation to cross-site external testing. As shown in Table 3 and Fig. 4, HAE-BUS yields the smallest AUC Gap in both transfer settings, with 6.84% for NDTH-to-BUSBRA and 4.39% for BUSBRA-to-NDTH. This suggests that HAE-BUS is less sensitive to site-level acquisition differences and maintains more stable performance under cross-site external testing.

5.5. Leave-one-device-out evaluation

To further evaluate robustness to device-level acquisition shifts, we conducted a leave-one-device-out (LODO) evaluation on BUSBRA. In each split, one ultrasound device was held out exclusively for testing, while the remaining devices were used for training and validation. The held-out device was not used for training, environment construction, hyperparameter selection, or threshold determination. Therefore, this protocol provides a stricter unseen-device evaluation than random intra-dataset splitting.

Table 4
Performance comparison under leave-one-device-out evaluation on BUSBRA.

Method	$\overline{\text{AUC}}$	$\overline{\text{ACC}}$	Worst AUC↑	AUC Gap↓
MixStyle	80.37 ± 2.20	71.56 ± 4.16	72.53 ± 4.12	7.84 ± 3.64
VREx	77.60 ± 3.98	68.64 ± 3.65	71.08 ± 4.36	6.52 ± 2.70
IRMv1	80.57 ± 2.40	72.72 ± 3.94	72.78 ± 2.62	7.80 ± 0.64
PLDG	81.99 ± 2.41	72.56 ± 3.79	75.34 ± 4.48	6.65 ± 4.83
CFFDL	80.73 ± 2.21	70.83 ± 4.79	72.80 ± 3.47	7.92 ± 1.65
HAE-BUS (Ours)	**82.30 ± 1.80**	**74.18 ± 1.68**	**77.53 ± 3.98**	**4.77 ± 3.65**

As shown in Table 4, all methods show lower performance under LODO than under random intra-dataset evaluation, indicating that device-level acquisition shift is challenging. HAE-BUS achieves the best overall performance, with the highest AUC of 82.30% and ACC of 74.18%, showing a 0.31% AUC advantage over PLDG and a 1.46% ACC advantage over IRMv1. It also achieves the highest Worst AUC of 77.53% and the smallest AUC Gap of 4.77%, indicating more stable performance across unseen devices.

The LODO results are lower than the NDTH→BUSBRA cross-site external validation results because cross-site external testing evaluates the mixed BUSBRA target cohort, whereas LODO evaluates each completely unseen device separately. Thus, LODO is more sensitive to device-specific acquisition characteristics, sample-size differences, and benign/malignant composition. Overall, these results suggest that device-level shift can be more challenging than site/cohort-level average transfer, while HAE-BUS still provides better relative robustness under this stricter acquisition-factor-based validation protocol.

Table 5
LOEO evaluation results across different held-out environments.

Method	Performance metrics				
	LOEO E_0	LOEO E_1	LOEO E_2	$\overline{\text{AUC}}$	$\overline{\text{ACC}}$
Random	79.84 ± 5.50	81.09 ± 5.95	84.09 ± 6.58	81.67 ± 5.88	77.31 ± 7.27
Img-KMeans	82.58 ± 6.98	89.55 ± 6.07	85.37 ± 5.68	85.83 ± 6.41	74.82 ± 8.36
Des-KMeans	83.48 ± 10.23	85.96 ± 5.16	85.51 ± 6.26	84.99 ± 4.62	77.07 ± 5.78
Metadata	82.46 ± 3.45	89.65 ± 6.75	81.44 ± 7.75	84.52 ± 2.62	74.73 ± 1.86
PRISM(Ours)	**84.51 ± 6.87**	**89.69 ± 6.59**	**88.85 ± 4.35**	**87.69 ± 6.07**	**80.36 ± 7.30**

Table 6
Influence of environment partitioning on the performance of ERM and DRO.

Partition	Training	AUC	Worst↑	Gap↓
Random	ERM	84.30 ± 1.88	76.46 ± 5.60	13.75 ± 6.27
	DRO	86.14 ± 2.71	80.93 ± 3.16	10.29 ± 3.62
Img-KMeans	ERM	87.48 ± 3.27	81.08 ± 6.64	10.33 ± 5.66
	DRO	83.99 ± 3.96	77.68 ± 6.13	11.94 ± 5.11
Des-KMeans	ERM	86.72 ± 1.99	78.18 ± 3.72	8.54 ± 2.61
	DRO	87.49 ± 1.64	79.83 ± 2.80	7.65 ± 2.35
Metadata	ERM	86.93 ± 2.48	75.69 ± 1.53	11.23 ± 3.02
	DRO	86.83 ± 2.51	77.29 ± 2.36	9.54 ± 1.56
PRISM(Ours)	ERM	86.52 ± 3.47	82.25 ± 6.00	7.83 ± 4.65
	DRO	**88.84 ± 2.10**	**85.47 ± 3.12**	**6.39 ± 5.68**

5.6. Influence of partitioning strategy on held-out generalization

To examine how environment construction affects cross-environment robustness, we perform a leave-one-environment-out (LOEO) evaluation, where one inferred environment is held out for testing and the remaining two are used for training. In Table 5, LOEO E_k denotes

Table 7
Results of ablation experiments under different component combinations. The column "4.4" indicates whether the Cross-Environment Task Composition with Adversarial Sampling module in Section 4.4 is used.

Method	Modules		Performance metrics			
	Partition	4.4	AUC	ACC	SEN	SPE
V1			85.45 ± 2.99	80.85 ± 3.24	59.71 ± 6.68	90.76 ± 3.66
V2	PRISM		88.56 ± 2.13	83.58 ± 1.05	70.43 ± 3.64	89.77 ± 1.04
V3	Random	✓	86.28 ± 3.62	82.72 ± 2.16	70.29 ± 7.01	88.55 ± 2.47
V4	Img-KMeans	✓	87.99 ± 3.20	83.29 ± 2.43	71.18 ± 5.94	88.97 ± 1.46
V5	Des-KMeans	✓	89.79 ± 1.90	84.60 ± 1.64	74.41 ± 3.54	89.38 ± 1.80
V6	Metadata	✓	88.94 ± 2.55	84.23 ± 1.68	75.88 ± 9.34	88.14 ± 6.68
Ours	PRISM	✓	**91.79 ± 1.05**	**87.88 ± 0.95**	**80.78 ± 5.62**	**91.19 ± 2.31**

the AUC obtained when environment E_k is held out. The results are reported in Table 5.

The compared partitions include Random assignment, image-feature K-means (Img-KMeans), descriptor-embedding K-means (Des-KMeans), scanner/device metadata grouping (Metadata), and the proposed prototype-guided MLLM assignment strategy (PRISM).

Several clear trends can be observed. Random partitioning yields relatively modest held-out performance, with an average AUC of 81.67% and an average ACC of 77.31%, indicating that arbitrary grouping provides only limited support for generalization under explicit environment exclusion. Img-KMeans improves the average AUC to 85.83%, suggesting that clustering in image-feature space captures part of the underlying shift structure. However, its average ACC drops to 74.82%, and the gap across held-out environments remains noticeable, which implies that feature-space partitioning alone does not consistently translate into stable decision boundaries. Des-KMeans obtains an average AUC of 84.99% and an average ACC of 77.07%, while Metadata achieves 84.52% AUC and 74.73% ACC. Both methods outperform Random partitioning in AUC but remain lower than PRISM, suggesting that descriptor clustering and scanner metadata capture partial acquisition heterogeneity but are less effective than PRISM-guided environment construction.

In contrast, PRISM achieves the best results on all three held-out environments, reaching 84.51% on LOEO E_0, 89.69% on LOEO E_1, and 88.85% on LOEO E_2. It also delivers the highest average AUC of 87.69% and the highest average ACC of 80.36%. Compared with all alternative partitioning strategies, PRISM provides stronger held-out generalization, suggesting that instruction-following partitioning better captures transferable acquisition-style variation.

5.7. Influence of environment partitioning on ERM and DRO

To examine how environment construction influences robust optimization, we compare ERM and DRO under different partitioning strategies. Results are reported in Table 6, where "Worst" denotes the worst-group performance and "Gap" measures the discrepancy between average and worst-group performance.

The benefit of DRO varies markedly with the partition quality. Under Random partitioning, DRO improves average AUC, raises worst-group performance, and reduces the gap, suggesting that even simple grouping can provide some useful structure for robust optimization. In contrast, under Img-KMeans, DRO lowers both average AUC and worst-group performance while enlarging the gap, indicating that image-feature clustering does not always produce groups that are well aligned with robust training. For Des-KMeans, DRO improves AUC/Worst from 86.72%/78.18% to 87.49%/79.83%, while reducing Gap from 8.54% to 7.65%. For Metadata, DRO slightly lowers AUC but improves Worst from 75.69% to 77.29% and reduces Gap from 11.23% to 9.54%.

PRISM yields the most favorable results and also shows the clearest gain from DRO. With ERM, PRISM already achieves strong average and worst-group performance with a relatively small gap. Replacing ERM with DRO further improves AUC from 86.52% to 88.84%, raises Worst from 82.25% to 85.47%, and reduces Gap from 7.83% to 6.39%. Overall, these results show that DRO is most effective when the induced environments better capture transferable cross-environment variation.

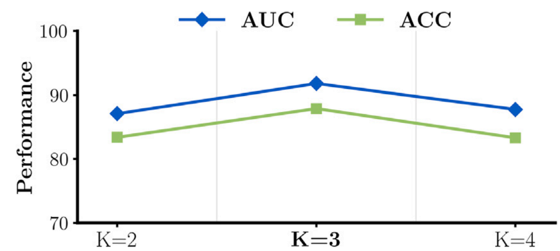


Fig. 5. Influence of the number of environments on model performance.

5.8. Ablation study

We conduct ablation experiments on the BUSBRA dataset to examine the contributions of the proposed PRISM-based environment partitioning and the Cross-Environment Task Composition with Adversarial Sampling module in Section 4.4. V1 removes both Instruction-Following Environment Partitioning and Section 4.4 module; V2 uses only PRISM-based partitions without Section 4.4 module; V3–V6 retain Section 4.4 module but replace PRISM with random partitioning, image-feature K-means, MLLM descriptor-embedding K-means, and scanner-metadata-based partitioning, respectively. Ours is the full HAE-BUS framework with both PRISM-based partitioning and stable learning.

The full model attains an AUC of 91.79% and ACC of 87.88%. V1, without either module, reaches 85.45% AUC and 80.85% ACC, reflecting limited robustness to cross-environment shifts under plain ERM. V2, which keeps partitions but omits stability, records 88.56% AUC and 83.58% ACC, indicating that semantic partitions alone – without adversarial, cross-partition composition – tend to yield conservative decision boundaries. Replacing PRISM with Random, Img-KMeans, Des-KMeans, or Metadata partitions reduces performance to 86.28%/82.72%, 87.99%/83.29%, 89.79%/84.60%, and 88.94%/84.23% in AUC/ACC, respectively. These results support that PRISM-guided pathology-excluded partitioning and cross-environment stable learning are both important for the final performance (see Table 7).

5.9. Influence of environment partition number

HAE-BUS integrates instruction-following partitioning and cross-environment task composition (Sections 4.3 and 4.4). We vary the number of environments $K \in \{2, 3, 4\}$ on BUSBRA; results are reported in Fig. 5.

$K = 3$ achieves the best performance, with 91.79% AUC and 87.88% ACC, and higher sensitivity of 80.78% at similar specificity of 91.19%. This setting provides sufficient granularity while retaining adequate samples per environment; With $K = 2$, performance drops to 87.08% AUC and 83.36% ACC. The coarser split may mix heterogeneous styles, increasing specificity to 92.29% but reducing sensitivity to 64.31%, which weakens cross-environment alignment; With $K = 4$, AUC and ACC are 87.75% and 83.08%, respectively. Over-fragmentation may reduce per-environment support and semantic coherence, undermining adversarial composition and generalization.

To further support this explanation, we analyze environment fragmentation under different K values in Fig. 6. When K increases from 2 to 4, the minimum number of samples per environment decreases from 243 to 130, indicating reduced per-environment sample support. Meanwhile, the malignant-ratio range increases from 0.010 to 0.070, suggesting larger class-composition fluctuation across environments. These results indicate that a larger K may lead to over-fragmented and less stable environment groups.

Overall, $K = 3$ offers the most favorable trade-off between informative style separation and stable optimization.

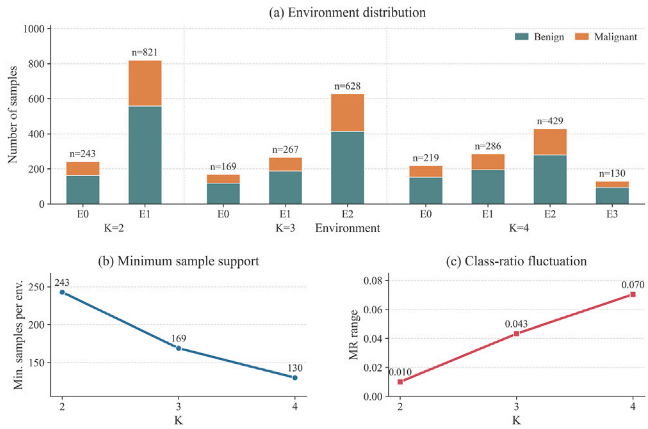


Fig. 6. Fragmentation analysis under different numbers of environments.

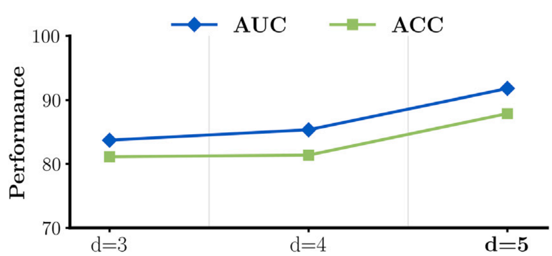


Fig. 7. Influence of different style-description settings.

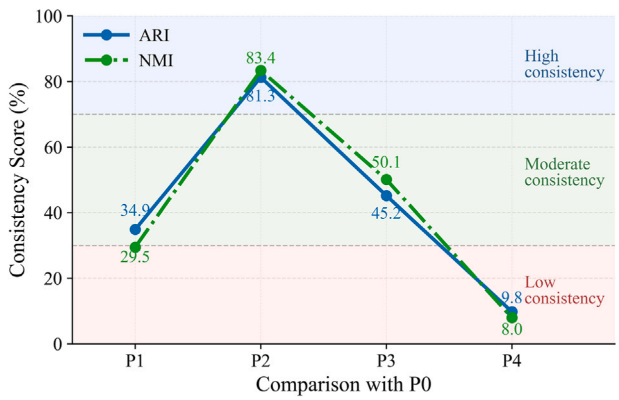


Fig. 8. Comparison of environment assignment consistency across different prompt variants.

5.10. Influence of style description dimensionality

To assess how description richness affects HAE-BUS, we perform a feature-ablation study within the instruction-following partitioning pipeline by varying the number of PRISM style dimensions $d \in \{3, 4, 5\}$ on the BUSBRA dataset. The full configuration uses $d = 5$ dimensions—`image quality`, `visual style`, `background texture`, `artifact presence`, and `machine signature`. Two reduced configurations are considered: the $d = 4$ setting removes `visual style`, and the $d = 3$ setting further removes `image quality`. Results are shown in Fig. 7.

The $d = 5$ configuration achieves the strongest performance, with an AUC of 91.79% and ACC of 87.88%. Reducing to $d = 4$ yields an AUC of 85.35% and ACC of 81.39%; further reducing to $d = 3$ yields an AUC of 83.72% and ACC of 81.13%. Sensitivity follows the same trend (80.78% → 62.35% → 65.10%), while specificity is relatively close (91.19% → 90.28% → 88.62%). These results indicate that `visual style` and `image quality` may provide complementary, pathology-excluded cues for forming coherent partitions and exposing meaningful cross-environment shifts; without them, the resulting partitions tend to be less informative, leading the model to adopt a more conservative decision boundary.

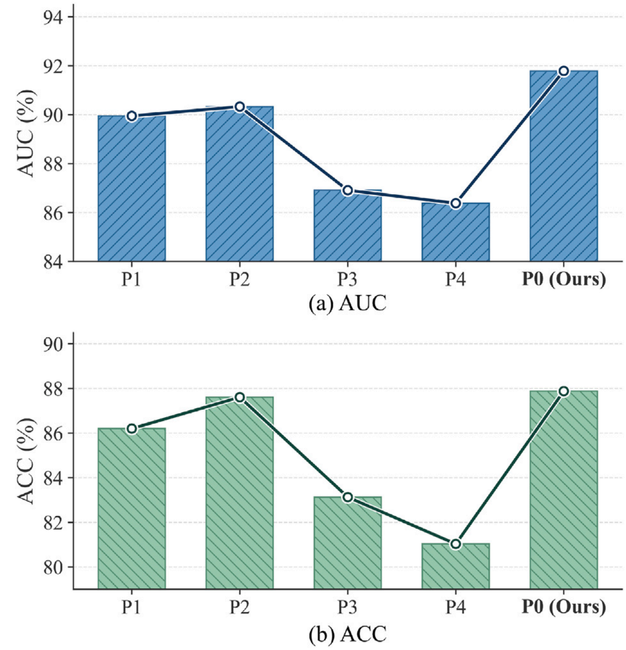


Fig. 9. Comparison of diagnostic performance across different prompt variants.

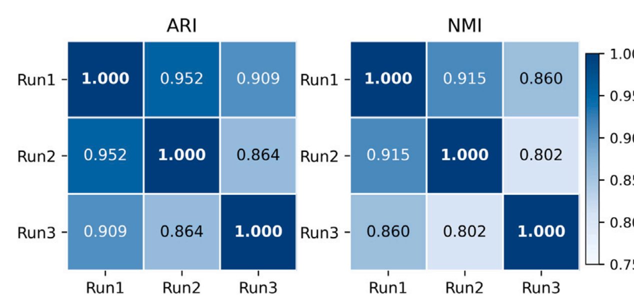


Fig. 10. Repeated-run consistency of MLLM-derived environment assignments on BUSBRA.

5.11. Prompt sensitivity analysis

To evaluate whether the proposed environment construction remains stable under prompt variations, we conducted a prompt sensitivity analysis on the BUSBRA dataset. The default prompt used in the main experiments is denoted as P0, and four variants were designed to evaluate semantic paraphrasing (P1), descriptor-field order perturbation (P2), weakened pathology-exclusion constraints (P3), and free-text prompting without the structured JSON schema (P4). The complete prompt templates are provided in the Supplementary Material.

Prompt sensitivity was evaluated from both environment consistency and downstream performance. Using P0 as the reference, P2 showed the highest agreement with P0 (ARI/NMI: 81.3%/83.4%), whereas P4 showed the lowest agreement (9.8%/8.0%), indicating that field-order perturbation has limited influence under structured prompting, while free-text prompting reduces reproducibility (see Fig. 8). Consistently, P0 achieved the best diagnostic performance (AUC/ACC:

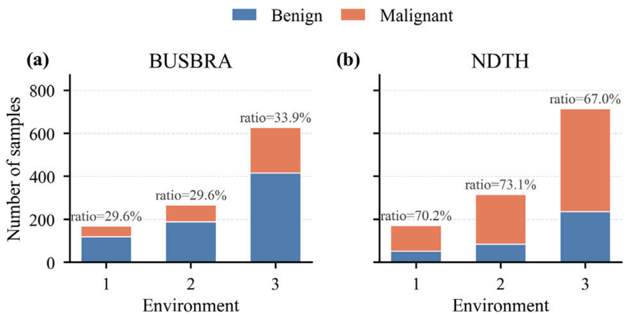


Fig. 11. Class distribution across inferred environments.

Table 8
Quantitative pathology-leakage analysis of MLLM-derived environments with and without lesion masking. All metrics are reported as percentages (%).

Dataset	Mask	Des-AUC	Des-ACC	Label-AUC	Label-ACC	Cramér's V	NMI
BUSBRA	×	57.46 ± 2.58	61.97 ± 1.20	52.07 ± 4.24	54.43 ± 3.70	6.06	0.22
BUSBRA	√	51.08 ± 0.66	51.55 ± 1.02	48.89 ± 2.29	49.44 ± 5.88	2.59	0.04
NDTH	×	70.03 ± 0.70	67.63 ± 1.17	58.43 ± 3.08	53.91 ± 2.57	15.72	1.71
NDTH	√	52.33 ± 1.77	55.02 ± 2.05	58.26 ± 4.11	53.74 ± 3.19	15.04	1.55

91.79%/87.88%), and P2 remained comparable (90.33%/87.61%), whereas P3 and P4 showed larger drops (see Fig. 9).

We further evaluated repeated-run consistency under the same prompt setting by repeating the complete MLLM-based environment assignment three times on BUSBRA. As shown in Fig. 10, the three runs showed high pairwise consistency, with mean ARI and NMI values of 0.908 and 0.859, respectively. These results suggest that fixed structured prompts and strict pathology-exclusion constraints are important for stable and reproducible environment construction and downstream performance.

5.12. Pathology-leakage analysis

We evaluated pathology leakage using descriptor-to-label prediction, environment-label-only prediction, Cramér's V, and NMI. Des-AUC/ACC measure whether MLLM-generated style descriptors predict benign/malignant labels, while Label-AUC/ACC evaluate whether inferred environment labels alone encode pathology labels. We further evaluated these indicators under both lesion-masked and unmasked settings.

As shown in Table 8, removing lesion masking increases descriptor-level leakage, especially on NDTH, where Des-AUC/ACC reach 70.03% /67.63%. With lesion masking, descriptor-to-label prediction is close to random on both datasets, with Des-AUC values of 51.08% on BUSBRA and 52.33% on NDTH. Meanwhile, the environment-label indicators remain low or change only slightly, suggesting that the inferred environments are not strong benign/malignant proxies.

Together with the class-distribution visualization in Fig. 11, these results indicate that lesion masking does not amplify pathology leakage. Instead, it suppresses lesion-related information during offline descriptor generation and encourages the MLLM to focus on non-diagnostic acquisition-style factors.

5.13. Qualitative analysis

We qualitatively examine three PRISM-partitioned environments using representative images and MLLM-generated style descriptions (Fig. 12). Within each environment, samples show consistent stylistic traits, and across environments the separation reflects acquisition appearance rather than pathology.

Environment 1 features cleaner exposure, wider dynamic range, smoother backgrounds, and minimal artifacts, consistent with stable device settings. Environment 2 is darker and grainier, with limited dynamic range and more shadowing/overlays, indicating angle- or

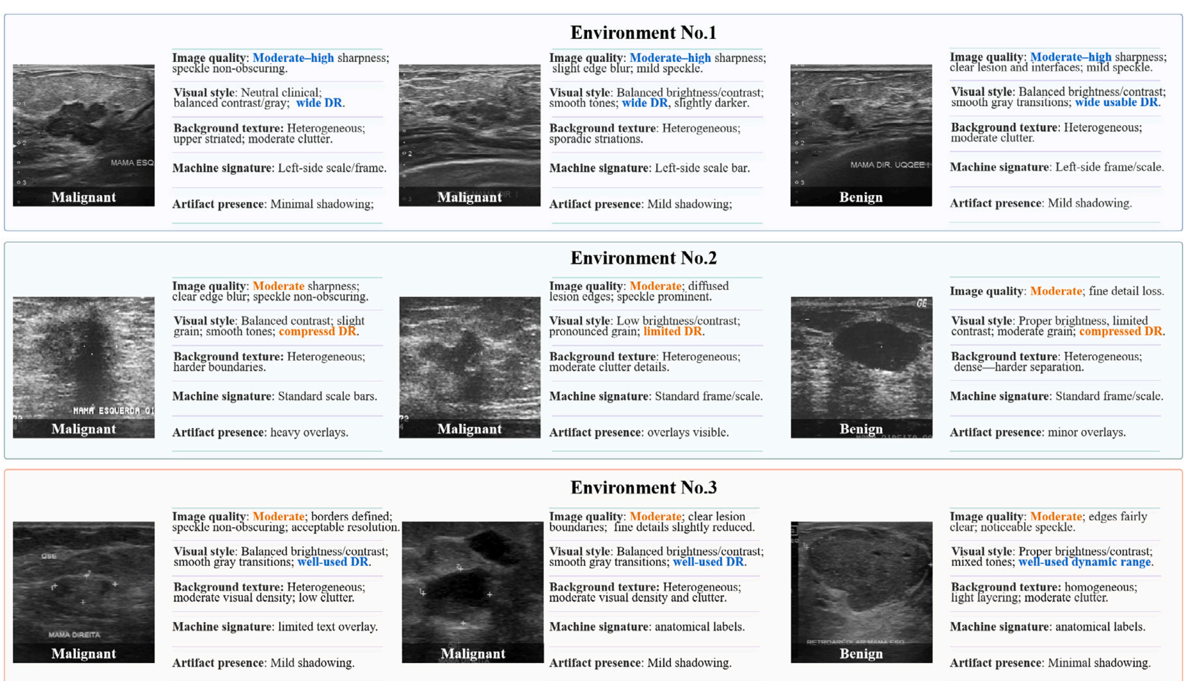


Fig. 12. Qualitative visualization of PRISM-based environment partitioning ($K = 3$). Each row corresponds to one environment (Env No. 1–No. 3), showing representative malignant and benign cases with MLLM-generated, pathology-excluded style descriptions across five dimensions (`image quality`, `visual style`, `background texture`, `machine signature`, `artifact presence`). Env No. 1 features moderate–high sharpness, wide dynamic range, and relatively clean backgrounds; Env No. 2 shows darker exposure, compressed dynamic range, heavier grain, and more overlays/shadowing; Env No. 3 lies between them, with moderate quality and well-used dynamic range. The consistent traits within each row and clear contrasts across rows indicate that PRISM yields semantically coherent partitions driven by acquisition style rather than pathology.

operator-dependent variability. Environment 3 lies between the two, showing moderate quality with smoother gray transitions, mild artifacts, and more balanced exposure.

Overall, the clear intra-environment coherence and inter-environment contrasts indicate that PRISM produces semantically meaningful, pathology-excluded partitions that support subsequent cross-environment stable learning.

6. Conclusion

We propose **HAE-BUS**, a representation learning framework for breast ultrasound diagnosis across heterogeneous acquisition environments, which integrates PRISM-based pathology-excluded style reasoning, instruction-following environment partitioning, and adversarial cross-environment composition to suppress acquisition-style bias while preserving pathology cues. By leveraging multimodal LLMs to generate structured style descriptions, HAE-BUS constructs interpretable and class-balanced environments without relying on metadata, providing an auditable basis for stable learning. Experiments on BUSBRA and NDTH show favorable AUC and accuracy, together with improved stability under cross-site external and device-level evaluation. Ablation and parameter analyses further confirm the effectiveness of PRISM-guided partitions and cross-environment task composition, suggesting that the proposed framework offers a promising direction for robust and interpretable breast ultrasound diagnosis.

CRediT authorship contribution statement

Huanjun Wang: Writing – original draft, Visualization, Validation, Project administration, Methodology. **Shukang Zhang:** Writing – review & editing, Formal analysis. **Wentao Kong:** Resources. **Wei Shao:** Writing – review & editing, Methodology. **Daoqiang Zhang:** Writing – review & editing, Resources. **Peng Wan:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Analysis of different MLLM backbones

We additionally compare several MLLM backbones for PRISM-style description generation on the BUSBRA dataset. This experiment aims to evaluate the effect of the description generator on downstream performance. For a fair comparison, the prompt template, the number of environments, and the downstream training protocol are kept unchanged, and only the MLLM backbone is varied. Results are reported in Table A.9.

The results show that different MLLM backbones lead to moderate performance differences. LLaVA-Med gives the lowest results among the compared models. InternVL2 and Qwen-VL both improve over LLaVA-Med in terms of AUC and ACC, indicating that improved description quality is beneficial to downstream environment partitioning. ChatGPT-4o achieves the best overall performance, obtaining the highest AUC, ACC, and SEN, while Qwen-VL achieves the highest SPE.

Overall, this supplementary comparison shows that downstream performance is influenced by the quality of PRISM-style descriptions, while the proposed framework remains applicable across different MLLM backbones.

Appendix B. Supplementary data

Supplementary material related to this article is provided as a separate supplementary file.

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.compmedimag.2026.102799.

Table A.9
Comparison of different MLLMs for PRISM-style description generation.

Method	Performance metrics			
	AUC	ACC	SEN	SPE
LLaVA-Med (Li et al., 2023)	85.21 ± 0.85	79.70 ± 4.04	70.59 ± 9.20	84.04 ± 9.48
InternVL2 (Chen et al., 2024)	88.22 ± 2.58	82.89 ± 2.32	72.55 ± 9.71	87.71 ± 7.12
Qwen-VL (Bai et al., 2023)	88.29 ± 1.89	83.13 ± 2.52	74.51 ± 5.18	**92.66** ± **1.53**
ChatGPT(Ours)	**91.79** ± **1.05**	**87.88** ± **0.95**	**80.78** ± **5.62**	91.19 ± 2.31

Data availability

The data that has been used is confidential.

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